MODULE 16 · DOMAIN 3 - INTEGRATION

Accuracy, Latency and Observability at Scale

Which latency levers are free and which cost accuracy, how to justify a configuration with measurements, and how to trace a request across agents and tools.

DOMAIN 3 of 7 EXAM WEIGHT 19% MODULE 16 of 38

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Make it right, then make it fast. And measure both, every time.

Three different latencies

MEASURE	WHAT IT CAPTURES	WHO FEELS IT
Time to first token	The wait before anything appears	A user watching a chat
Output speed	Tokens per second once it starts	Long answers and documents
End-to-end time	Every step, tool and model call added up	Agents, pipelines, batch jobs

ON THE EXAM

Name which latency the requirement means, and at which percentile, before choosing a lever.

Latency levers: free wins and real trades

USUALLY FREE IN QUALITY

Do these first

- Cache the stable prefix
- Stream the response
- Run independent tool calls in parallel
- Remove irrelevant context
- Route easy requests to a lighter tier, if tested

TRADE ACCURACY FOR SPEED

Measure before keeping

- A lighter tier for everything
- Lower effort, less thinking
- Fewer retrieved chunks, no reranker
- Shorter outputs, tight max_tokens
- Trimming examples or instructions

THE TIEBREAKER

Exhaust the free wins first. Every trade needs an evaluation result to justify it.

Justifying a configuration

01

Get it right

No speed constraints yet.

02

Baseline

Accuracy and latency on the eval set.

03

Change one lever

Free wins first.

04

Re-measure both

Same set, same percentile.

05

Keep or revert

Above the floor and faster, or undo.

ON THE EXAM

A configuration is justified by numbers on a representative set, not by a demo or a benchmark table.

Speed options, October 2026

OPTION	WHAT TO KNOW
Haiku 5.5	Fastest tier; effort is its main speed control
Effort	Lower effort, faster and cheaper; temperature is not the lever
Streaming	Improves perceived responsiveness, not total time
Prompt caching	Faster first token on long, repeated prefixes
Fast mode, research preview	Up to 2.5x output speed on Opus, at twice the price
Fast mode limits	Claude API only; not first-token speed; no Batch

ON THE EXAM

Fast mode speeds up output, not the wait for the first token. Read which latency the brief means.

What an LLM system must record

- **Versions:** model, prompt template, tool set, retrieval index, configuration.
- **Per step:** tokens in, out and cached; latency; tool calls and their duration.
- **Evidence:** IDs of retrieved chunks and documents, not just the final answer.
- **Identifiers:** correlation ID, plus each platform request ID.
- **Redaction:** personal and secret fields masked before storage.

> ON THE EXAM
>
> If you cannot reconstruct why an answer was given, you cannot fix it, defend it, or audit it.

One trace across agents and tools

Orchestrator span

Model and prompt version
Tokens and latency

Subagent spans

Same ID, own steps
Hand-offs recorded

Correlation ID

Created at the front door
Passed in every call

One ID, every hop

Tool and MCP spans

Arguments, redacted
Result and duration

Platform IDs

API request-id header
MCP trace context in _meta

THE PICTURE

A parcel tracking number that follows the parcel across every courier.

Monitoring at scale

WATCH	BECAUSE
p50 and p95 latency per step	Averages hide the slow tail users remember
Tokens, cost and cache hit rate	A broken cache shows up here first
Errors, timeouts and 429s	Rate limits and outages, per model and tool
Quality on sampled live traffic	Latency can be fine while answers drift
Changes marked on every chart	So shifts line up with deployments

ON THE EXAM

Dashboards show the symptom. Traces explain it.

> Quillon's portal assistant must show its first token within 2 seconds at p95. It measures 3.4 seconds, while scoring 95% against a 92% accuracy floor.
>
> Every request sends the same 18,000-token system prompt of policies, and responses are returned only when complete.

What should Corvane try first?

A Cache the stable system prompt and stream responses, then re-measure

B Move every request to the lightest tier straight away

C Cut the system prompt to half its length

D Turn on fast mode for the current model

◆ PAUSE AND DECIDE

What should Corvane try first?

A Cache the stable system prompt and stream responses, then re-measure

Two free wins aimed at first-token time.

B Move every request to the lightest tier straight away

A trade, untested against the 92% floor.

C Cut the system prompt to half its length

Removes policy the answers depend on.

D Turn on fast mode for the current model

Speeds output, not the first token. A preview, at twice the price.

THE SIGNAL

A first-token target with a long, stable prefix: cache it and stream. Free wins first.

> Tallyhawk's support agent passes 50 retrieved chunks to the model on every question. Tracing shows most of its latency is the model reading them.
>
> On the evaluation set, accuracy is unchanged with 15 chunks, and drops with 5.

Which change should Corvane make?

A Remove the reranker to save its processing time

B Move the agent to the lightest model tier

C Pass the smallest number of chunks the evaluation tested

D Pass the smaller number of chunks that the evaluation shows keeps accuracy

◆ PAUSE AND DECIDE

Which change should Corvane make?

A Remove the reranker to save its processing time

Not where the time goes, and it protects accuracy at fewer chunks.

B Move the agent to the lightest model tier

Untested trade. The trace points elsewhere.

C Pass the smallest number of chunks the evaluation tested

Five chunks lost accuracy. Faster, and worse.

D Pass the smaller number of chunks that the evaluation shows keeps accuracy

The trace found the cost; the eval found the safe setting.

THE SIGNAL

Let the trace find the time and the evaluation find the safe setting. Then change that one lever.

> A Mereside adviser complains that agent assist gave a wrong answer on a call last Tuesday. Corvane cannot reconstruct why.
>
> Logs hold only the final suggestions, with no record of retrieved policy sections, prompt version or model version.

What should Corvane add?

A Keep the final suggestions for longer, with exact timestamps

B A trace per request with a correlation ID, model and prompt versions, retrieved chunk IDs and per-step latency, with personal data redacted

C Ask Claude to explain its reasoning at the end of every suggestion

D Store every full prompt, including call transcripts, indefinitely

◆ PAUSE AND DECIDE

What should Corvane add?

A Keep the final suggestions for longer, with exact timestamps

More of what already failed to explain anything.

B A trace per request with a correlation ID, model and prompt versions, retrieved chunk IDs and per-step latency, with personal data redacted

Reconstructs the inputs behind any answer, safely.

C Ask Claude to explain its reasoning at the end of every suggestion

A story about the answer, not a record of its inputs.

D Store every full prompt, including call transcripts, indefinitely

A privacy problem, under UK GDPR, to solve a design one.

THE SIGNAL

Cannot explain an answer: record versions and evidence per request, redacted. Not more answers.

> Tallyhawk's fault-diagnosis system has an orchestrator, three subagents and several MCP servers. Each component writes its own logs.
>
> When a customer reports a slow diagnosis, engineers cannot tell which agent or tool call was slow for that conversation.

What should Corvane recommend?

A. Create a correlation ID per conversation and pass it through every agent, tool and MCP call, recording a span for each

B. Synchronise all server clocks and match logs by timestamp

C. Write every component's logs into one shared file

D. Have each subagent summarise its own work when it finishes

◆ PAUSE AND DECIDE

What should Corvane recommend?

A Create a correlation ID per conversation and pass it through every agent, tool and MCP call, recording a span for each

One ID joins every hop; spans show where the time went.

B Synchronise all server clocks and match logs by timestamp

Concurrent conversations overlap. Timestamps cannot separate them.

C Write every component's logs into one shared file

One pile, still unjoined.

D Have each subagent summarise its own work when it finishes

No timings, and written by the system being diagnosed.

THE SIGNAL

Many agents and tools, one request to explain: a correlation ID with a span per hop.

> Corvane proposes moving Quillon's routine portal questions to the Haiku tier at low effort, to cut latency and cost.
>
> Quillon's change board asks what evidence supports the change.

What evidence should Corvane present?

A Anthropic's published benchmark scores for both tiers

B A live demonstration answering ten typical questions

C The engineering team's judgement that the lighter tier is good enough

D Both configurations on the portal evaluation set: accuracy against the floor, and first-token time against the target

◆ PAUSE AND DECIDE

What evidence should Corvane present?

A Anthropic's published benchmark scores for both tiers

Not Quillon's questions, not Quillon's bar.

B A live demonstration answering ten typical questions

Ten easy cases prove little about the tail.

C The engineering team's judgement that the lighter tier is good enough

Opinion, where the board needs numbers.

D Both configurations on the portal evaluation set: accuracy against the floor, and first-token time against the target

Both requirements, measured on Quillon's own data.

THE SIGNAL

Justify a configuration with both requirements, measured on a representative set, side by side.

> Tallyhawk's support agent makes four independent account lookups, one after another, on most billing questions.
>
> Every request also resends a long, unchanging system prompt and tool list, uncached. Answer quality must not fall.

Which two changes cut latency without reducing answer quality? Choose two.

A Move all traffic to the lightest model tier

B Let the agent request the independent lookups in parallel

C Set effort to low on every request

D Pass only two retrieved chunks per question

E Cache the stable system prompt and tool list

◆ PAUSE AND DECIDE

Which two changes cut latency without reducing answer quality? Choose two.

A Move all traffic to the lightest model tier

A trade, risking the hard cases.

B Let the agent request the independent lookups in parallel

Four round trips become one. Same answers.

C Set effort to low on every request

Less reasoning on every case, hard ones included.

D Pass only two retrieved chunks per question

Faster, and likely to miss evidence.

E Cache the stable system prompt and tool list

Faster first token, identical output.

THE SIGNAL

Without reducing quality means free wins only: parallel calls and caching.

In five lines

- **Make it right, then fast.** Name which latency, at which percentile.
- **Free wins first:** caching, streaming, parallel calls, less irrelevant context.
- **Trades need evidence:** tier, effort, chunks, length. One lever at a time, both measured.
- **Record versions and evidence per request,** redacted, under one correlation ID.
- **Dashboards show the symptom; traces explain it.**

COMING UP

Domain 3 complete. Next, Domain 4: metrics and success criteria.